S-Series v4 Cabinets

Site Preparation and Readiness Guide

S-Series v4 – © 2025

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ADDITIONAL DOCUMENTATION MAY BE PROVIDED AS REQUIRED THROUGHOUT THE DELIVERY AND INSTALLATION PROCESS.

TECHNICAL SUPPORT SITE

Should you encounter any installation or operational issues with your product, check the pertinent section of this manual to see if the issue can be resolved by following the outlined procedures.

You can also email us at support@ddcsolutions.com for additional assistance.

REVISION HISTORY

Revision No.	Description	Date	Prepared By	Checked By	Approved By	Updates
00	DRAFT	10/30/24	CJM			
01	Init. Issue	1/17/25	CJM			
02	Update	2/19/25	CJM			Added cabinet dimensions
03	Update	3/7/25	CJM			Added CHW valve wiring requirements. Added requirements for DCIM Network connections.
04	Update	3/17/25	CJM			Included request for floorplan / DCIM “landing page”.
05	Update	6/10/25	CJM			Updated input power requirements for cabinet controller.
06	Update	9/18/25	CJM			Removed reference to Analytics Enclosures. Updated power requirements section. Corporate Logo update.

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Product Description

The DDC Solutions S-Series v4 Cabinet consists of a self-contained, water-cooled coil unit, supply and exhaust fans, integrated fire suppression, and control software in a cabinet enclosure that incorporates a standard 19" IT rack.

Data Hall Ambient Temperature Recommendations

ASHRAE (American Society of Heating, Refrigerating, and Air-Conditioning Engineers) recommends a temperature range in the data hall between 18° to 27°C (64.4° to 80.6°F). This metric is based on dry-bulb temperature. Wherever possible, relative humidity should be kept between 25% and 50%.

Chilled Water System

The DDC Solutions S-Series v4 Cabinet technology uses two aluminum chilled water coils as the primary source of heat exchange. The coils transfer heat from the source to the destination through a series of tubes, using chilled water provided by the customer's chilled water plant.

Accounting for and achieving proper water quality and chemistry is critical for all water-cooled applications. Do not charge coils until the entire system is thoroughly flushed of debris and particulate, and water chemistry is confirmed to be within the appropriate range for aluminum and other present materials by a qualified water treatment professional. A corrosion inhibitor for multi-metal systems must be used if fluid piping contains copper, brass, iron or other metals when used in conjunction with aluminum heat exchangers. All dissimilar metal joints within the system in which this product is being installed must be properly insulated from dielectric action.

This is a closed-loop system. It should be closed to the atmosphere.

Corrosion Inhibitor

A corrosion inhibitor for multi-metal systems must be used if fluid piping contains copper, brass, iron or other metals when used in conjunction with aluminum heat exchangers.

Glycol

Most glycols contain a corrosion inhibitor. The corrosion inhibitor must be verified to ensure it provides multi-metal protection and is safe for aluminum. It is important to note that the use of glycol in the system may have an impact on the overall performance as it relates to the heat abatement of the product.

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pH Requirements

Proper water chemistry is crucial for the proper operation of the system. pH MUST be maintained between 7.5 and 8.5. Water chemistry should be monitored weekly and adjusted as needed. Out of range pH is a common cause of early coil failure. Prior to charging the system, water chemistry MUST BE TESTED AND CONFIRMED TO BE WITHIN THE REQUIRED pH RANGE, and weekly upon commissioning. **Coil failure resulting from incorrect pH levels is not covered by the DDC Solutions or product manufacturer's warranty**.

Incoming Water Temperature:

- **Recommendation**: For optimal system performance, DDC Solutions recommends an incoming water temperature of 57°F (14°C)
- **Supported Range**: DDC Solutions S-Series Cabinets support incoming water temperatures between 40°F and 65°F (4.4°C and 18.3°C).
  Important considerations:
  - Incoming water temperatures MUST be above the dewpoint to avoid condensation. This is especially critical in potentially humid environments.
  - Water temperatures above the recommended setpoint may or will have an adverse impact on the overall performance as it relates to the heat abatement capacity of the product.

Chilled Water Coils & Valves:

Each cabinet is equipped with two chilled water coils, each controlled separately by electronic water flow valves. The control valves should be installed on the RETURN side.

- **Pressure and GPM Requirements**: Minimum of 18.2GPM at 20psig (36.4GPM/cabinet).
- **Supply and Return Connections:** Two chilled water supply and return connections are required for each cabinet, equipped with their own shut-off valves. Connection requirements are 1" NPT. Fittings and valves are included in the plumbing ship-loose kit included with each cabinet. Customers are responsible for providing their own hoses.
- **Control Wiring Requirements**: There are two electronically controlled flow valves – each valve requires one (1) network cable (CAT5e or CAT6) that is used to receive feedback from the valve, and one (1) 18/3 stranded, shielded security type cable to control and provide power to the valve. These cables are provided and installed by the customer/customer's install contractor. Extra length should be coiled at each end for final termination by the DDC commissioning team. If desired, customers may use the plumbing hose(s) for cable management. All wiring must be attached to the outside of any insulation or pipe wrap that may be added to the hose(s). We recommend using Velcro straps vs. zip ties wherever practical.
- **Coil Design Working Pressure**: 50psig (3.4bar)
- **Coil Design Overall Internal Volume (in cu in)**: 127.9
- **Chilled Water Strainers:**
  - **Site Level Recommendation**: A strainer of 120 to 140 Mesh is recommended
  - **Cabinet Coil Level**: A strainer of 60 Mesh is recommended at each coil

Row Length Considerations

- DDC recommends that the S-Series v4 Cabinets are installed in rows of 12-units or less

Dimensions, Weights and Service Clearances

- Dimensions (30″ Width):
  - 45U: 115″ H; 30″ W; 82″ D
  - 48U: 121″ H; 30″ W; 82″ D
  - 52U: 128″ H; 30″ W; 82″ D
- Dimensions (36″ Width)
  - 45U: 115″ H; 36″ W; 82″ D
  - 48U: 121″ H; 36″ W; 82″ D
  - 52U: 128″ H′ 36″ W; 82″ D
- Min. Service Clearance: 36″ Top; 48″ Front / Rear
- NEMA Design: NEMA 3R
- Weight: ~850lb

Power Requirements

Cabinet Control (Per Cabinet)

- **208VAC Controller Option**: One L15-30R (208V 3-ph 30A) connector on a dedicated circuit for each cabinet. The cabinet will come equipped with a top mounted 2′ whip, terminated with an L15-30P connector.

PDUs (IT Load)

- # of PDUs / Cabinet: FROM ORDER
- PDU Type: FROM ORDER
- Circuits Required: FROM ORDER

DCIM Network

The DDC DCIM Network is a private, separate network that is used to facilitate the operation and monitoring of the customer's cabinets. This network is comprised of customer-provided servers (physical or virtual) for both the DCIM application and database, customer managed << ROUTERS / SWITCHES >>, DDC provided analytics enclosure(s) and a customer provisioned and managed VPN access to support DDC's remote support. Cabinet to cabinet network connectivity (exclusive of the production IT network) is provided by the customer at the time of installation using Cat6E (or better) ethernet. Details and specifications are outlined below:

Application Server

- Operating System: Windows Server 2019 (or newer) – 64-bit
- CPU: Recommended 8 Cores
- RAM: Minimum of 16B (Recommend 64GB for optimum performance)
- NIC: 1GigE (or better)
- Storage: 4TB – PCIe SSD

Database Server

- Operating System: Windows Server 2019 (or newer) – 64-bit
- CPU: Recommended 8 Cores
- RAM: Minimum of 16B (Recommend 64GB for optimum performance)
- NIC: 1GigE (or better)
- Storage: 20TB – PCIe SSD
- SQL Software: SQL Server 2016 (or newer)

Floorplan / Data Center Layout Drawing

Part of the DCIM software provisioning / updates includes a landing page that looks like the actual data center. If you would like your DCIM site level page to reflect your cabinet layout, please provide a floorplan of the area in which the cabinets will be installed.

Switches / Routers

The DCIM network will require connection via a customer provided and managed (switch / router). These devices should meet the following minimum requirements:

- Enough ports to support two (2) connections per row
- Ports to match type of connection run to ends of each row (fiber or ethernet)

Cross Connect Requirements

To facilitate communication between the all the cabinet(s) and the DCIM network, the cabinets are wired in a "daisy chain" configuration, with a connection from the end cabinets of each row back to the customer managed switch / router. Requirements follow:

- Per Cabinet:
  - Two (2) SFPs (single mode fiber is preferred – GigE/Cat6 is also acceptable). These can be left on top of each cabinet. The DDC commissioning team will make the final connections.
  - Depending on the SFP type, one SM fiber or Cat6 ethernet. Wiring can be coiled at the top of the cabinet(s) with a 6' service loop. The DDC Commissioning Team will make the final connections.
- At ends of each row:
  - Single-mode fiber (or Cat6 or better Ethernet) connection will need to be extended from the cabinet to the customer provided managed switch / router. Connections can be coiled at each end with a service loop. The DDC Commissioning Team will make the final connections.

Customer External IT Network Connections

To meet NEMA-3R certification, the DDC S-Series cabinets are sealed at the factory prior to shipment. We understand that each customer has unique requirements for network connectivity into and out of the cabinet. At the top and bottom corner of each cabinet are 5" x 5" removable blanks that can be modified by the customer's installing contractor to meet their specific needs. To maintain the NEMA-3R certification, DDC Solutions recommends the use of watertight connectors when modifying the removable blanks.

Delivery Arrangements

Delivery within the United States or Canada is by air ride truck, with custom made shipping material (e.g. crates and/or pallets). International delivery normally involves air freight.

DDC Solutions will arrange for delivery directly to the customer's loading dock. The customer is responsible for the uncrating, inspection, relocation and assembly in the customer's data center, and removal/disposal of all packaging materials.

Pre-Delivery Considerations

The following should be considered prior to delivery:

- Weight capacities of the loading dock (if any)
- Service elevator requirements (if applicable)
- Length and thickness of any floor covering (if applicable)
- Availability of ramps, moving dollies, lifts, etc. to facilitate the safe unloading, uncrating, relocation, and installation of equipment
- Loading dock hours
- Special delivery requirements (e.g. short truck, lift gates, etc.)
- Special security or access requirements for DDC staff / drivers

The entire team at DDC Solutions thanks you for your trust and we look forward to working with you on this project. Please contact your project team at any time if there are questions or concerns.

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